IFAC World Congress 2023

DEPARTMENT OF ENGINEERING SCIENCE

UNIVERSITY OF OXFORD

Discrete Lagrangian Neural Networks with Automatic Symmetry Discovery

Yana Lishkova

P. Scherer, S. Ridderbusch, M. Jamnik, P. Liò, S. Ober-Blöbaum, C. Offen

Yokohama, Japan, July 2023



Outline

- **Background**
  - System dynamics and conservation laws
  - Discretization and integration schemes
- **Our approach for system identification**
- **Results**
  - Trajectory recreation and prediction
  - Energy and Symmetry preservation
  - Learning from noisy data
- **Future plans**

Mathematical model

$q(t)$ - generalized coordinates

$p(t)$ - generalized momenta, $p = \frac{\partial \mathcal{L}}{\partial \dot{q}}$

Kinetic energy $T^s(q, \dot{q}) = T^p(q, p)$

Potential energy $V^s(q, \dot{q}) = V^p(q, \dot{q})$

Hamiltonian approach

Hamiltonian: $H(q, \dot{q}) = T^p(q, p) + V^p(q, p)$

Equations of motion:

$$\dot{p} = -\frac{\partial H(q,p)}{\partial q}, \dot{q} = \frac{\partial H(q,p)}{\partial p}$$

Lagrangian approach

Lagrangian: $\mathcal{L}(q, \dot{q}) = T^s(q, \dot{q}) - V^s(q, \dot{q})$

Equations of motion:

$$\frac{\partial \mathcal{L}(q,\dot{q})}{\partial q} - \frac{d}{dt}\frac{\partial \mathcal{L}(q,\dot{q})}{\partial \dot{q}} = 0$$

*Cranmer, M., Greydanus, S., Hoyer, S., Battaglia, P.,Spergel, D., and Ho, S. (2020). Lagrangian neural networks. arXiv preprint arXiv:2003.04630.
**Greydanus, S., Dzamba, M., and Yosinski, J. (2019). Hamiltonian neural networks. Advances in neural information processing systems, 32.

Conservation laws

Noether's theorem links the **invariance (or "symmetry")** of a system to a conservation law in the motion. For example:

- **conservation of energy (time invariance)**

  $\frac{\partial H}{\partial t} = 0$ (rate of change of energy is zero)

- **momentum (translation invariance)**

  $\dot{p} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{\partial \mathcal{L}(q,\dot{q})}{\partial q} = 0$ (rate of change of momentum is zero)

- **angular momentum (rotational invariance)**



Emmy Noether

More generally a conserved quantity (symmetry) along the trajectory takes the form :

$$I(q,\dot{q}) \;=\; (M\,q + w)^T\,\nabla_{\dot{q}}\,\mathcal{L}(q,\dot{q})$$

We have achieved a neural network model which based on Lie Group theory can learn the conservation principle along from only configuration data.

***Lishkova, Y., Scherer, P., Ridderbusch, S., Jamnik, M., Liò, P., Ober-Blöbaum, S. and Offen C.** (2022) Discrete Lagrangian Neural Networks with Automatic Symmetry Discovery. arXiv preprint :2211.10830.
****Dierkes, E., Offen, C., Ober-Blöbaum, S. and Flaßkamp, K. (2023).** Hamiltonian Neural Networks with Automatic Symmetry Detection, preprint available online

Mathematical model

Hamiltonian approach

Equations of motion:

$$\dot{p} = -\frac{\partial H(q,p)}{\partial q}, \dot{q} = \frac{\partial H(q,p)}{\partial p}$$

Lagrangian approach

Equations of motion:

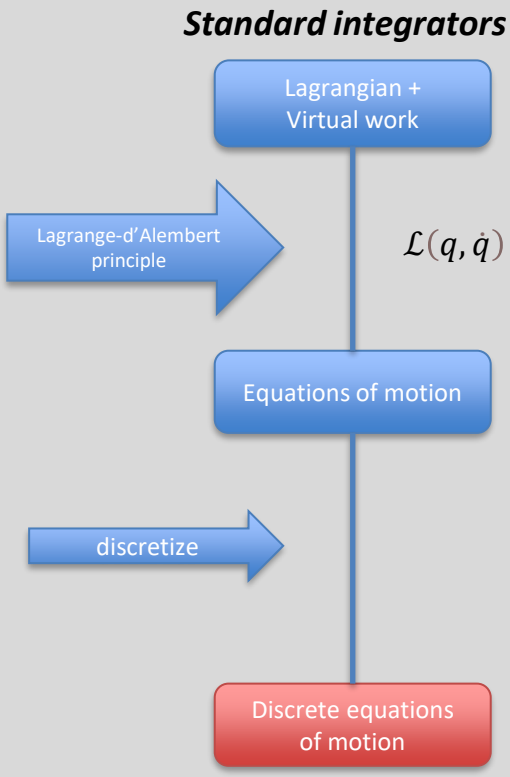
$$\frac{\partial \mathcal{L}(q,\dot{q})}{\partial q} - \frac{d}{dt}\frac{\partial \mathcal{L}(q,\dot{q})}{\partial \dot{q}} = 0$$

Solution options to obtain $q(t)$ at all times of interest:

- **Analytic solution**
  - rarely available
- **Standard numerical integrator**
  - focus on ***quantitative*** error*
  - non-satisfactory long-term accuracy**
- **Geometric numerical integrator**
  - focus on ***qualitative*** representation*
  - Accurate long-term integration

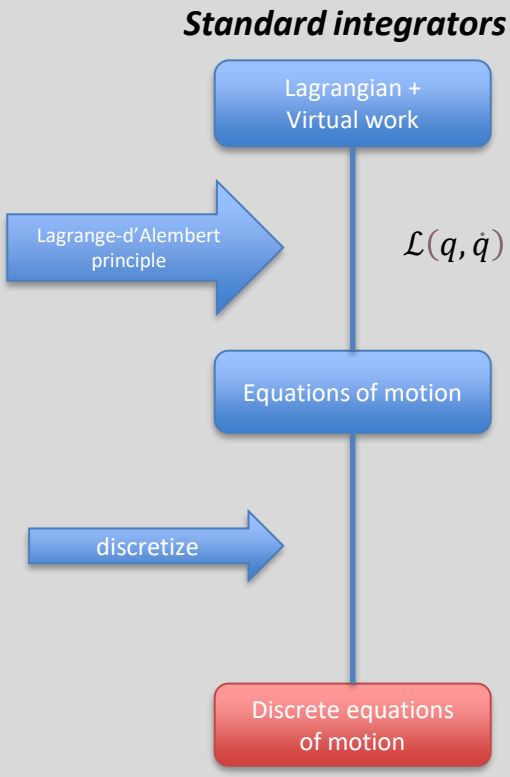
*Arieh Iserles. *A First Course in the Numerical Analysis of Differential Equations.* Cambridge Texts in Applied Mathematics. Cambridge University Press, 2 edition, 2008.
**Dina Razandralandy, Aziz Hamdouni, and Marx Chhay. A review of some geometric integrators. *Advanced Modeling and Simulation in Engineering Sciences*, 5(1):1-67, 2018.

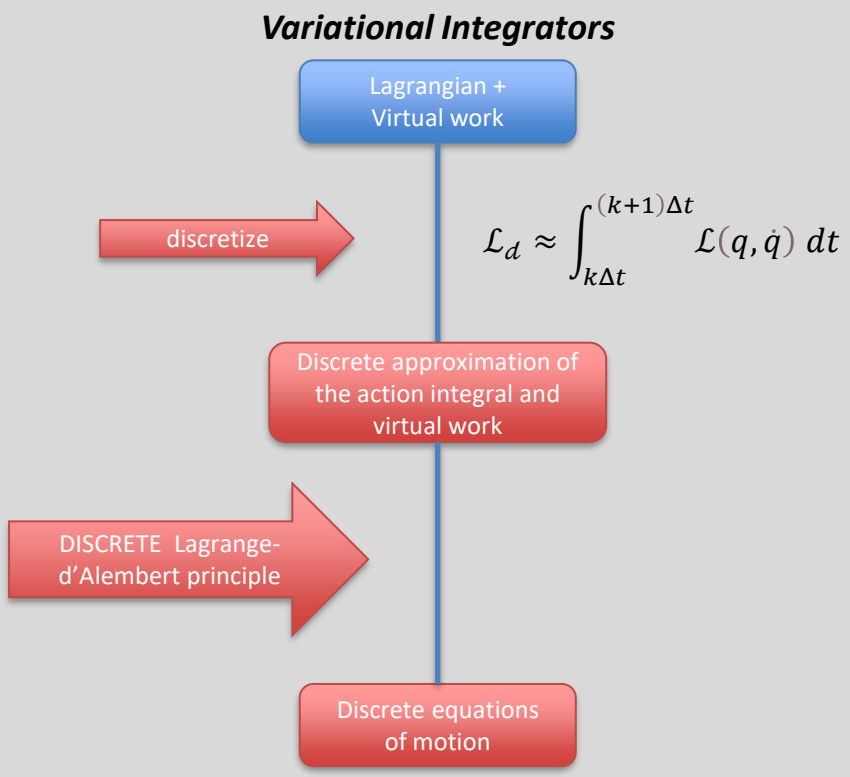
Variational integrators



*** J. E. Marsden and M. West,** "Discrete mechanics and variational integrators," *Acta Numerica*, vol. 10, no. 1, pp. 357–514, 2001

Variational integrators





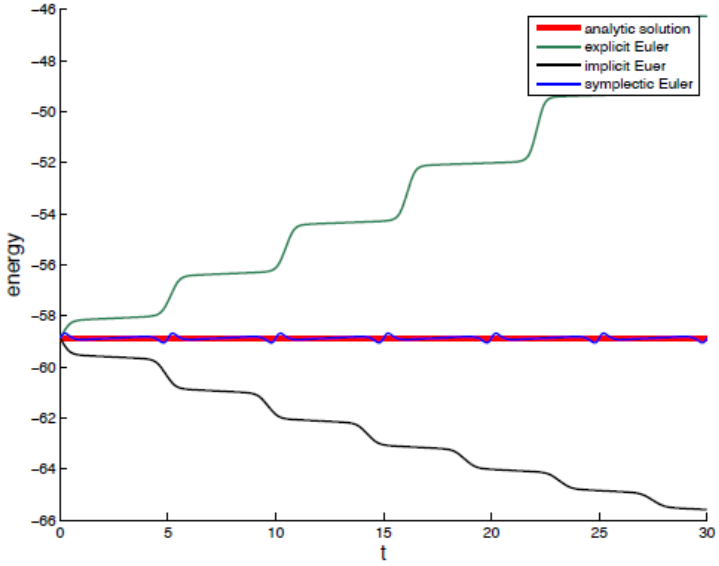
* **J. E. Marsden and M. West,** "Discrete mechanics and variational integrators," *Acta Numerica*, vol. 10, no. 1, pp. 357–514, 2001

Example of structure importance

UNIVERSITY OF OXFORD

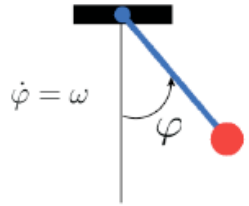
Symplektische Methoden in der optimalen Steuerung

Sina Ober-Blöbaum



Simulation of the pendulum

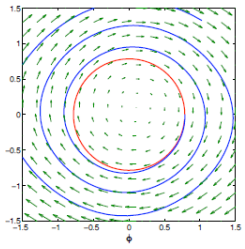
- pendulum: $\dot{\varphi} = \omega,\ \dot{\omega} = -\sin(\varphi)$
- $\dot{x} = f(x)$ with $x = (\varphi, \omega)$ and $f(x) = \begin{pmatrix} \omega \\ -\sin(\varphi) \end{pmatrix}$



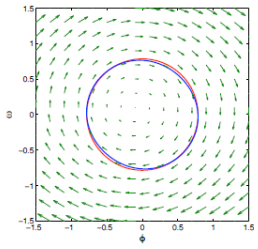
explicit Euler
$\varphi_{k+1} = \varphi_k + h\omega_k$
$\omega_{k+1} = \omega_k - h\sin\varphi_k$

symplectic Euler
$\varphi_{k+1} = \varphi_k + h\omega_k$
$\omega_{k+1} = \omega_k - h\sin\varphi_{k+1}$

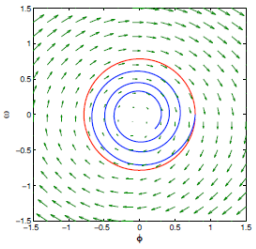
implicit Euler
$\varphi_{k+1} = \varphi_k + h\omega_{k+1}$
$\omega_{k+1} = \omega_k - h\sin\varphi_{k+1}$



explicit Euler



symplectic Euler



implicit Euler

Proposed methods

	LNN*	DLNN	SymDLNN
No restrictions on the expression of the Lagrangian being learnt	✗	✓	✓
Only configuration data needed	✗	✓	✓
Direct discrete Lagrangian identification *(reduces the discretisation errors preventing the amplification of the modelling error)*	n/a	✓	✓
Symmetry identification (*qualitative better solution and potentially faster learning)*	✗		✓
Symmetry preserving simulation *(conservation of energy and/or momenta of the system)*	✗	✓	✓
Incorporates a degeneracy cost term *(safeguards against unwanted solutions and against potential numerical issues in forward simulations)*	✗	✓	✓
Capability to manage noisy data	✓	✓	✓

***Cranmer, M., Greydanus, S., Hoyer, S., Battaglia, P.,Spergel, D., and Ho, S.** (2020). Lagrangian neural networks. arXiv preprint arXiv:2003.04630

Results

Example problems

Pendulum on a cart

$$\mathcal{L}_C(s, \phi, \dot{s}, \dot{\phi}) = \frac{1}{2}\left(\alpha\dot{\phi}^2 + 2\beta\cos(\phi)\,\dot{s}\dot{\phi} + \gamma\dot{s}^2\right) + D\cos(\phi)$$

with

$$\alpha = m_1 l^2,\ \beta = m_1 l,\ \gamma = m_2 + m_1,\ D = -m_1 g l,$$

$$m_1 = m_2 = l = 1$$

m_2 - mass of the cart, m_1- mass of the pendulum,

l - length of the pendulum, g - gravity acceleration

and translational symmetry

$$M = [0,0;\ 0,0],\ w = [1,0]$$

Kepler problem

$$\mathcal{L}_K(x, y, \dot{x}, \dot{y}) = \frac{1}{2}\left(\dot{x}^2 + \dot{y}^2\right) + \frac{G m_1 m_2}{\sqrt{x^2 + y^2}}$$

with

$$G = 6.673 \times 10^{-26},\ m_1 = 6 \times 10^{24},\ m_2 = 100$$

and rotational symmetry of the form

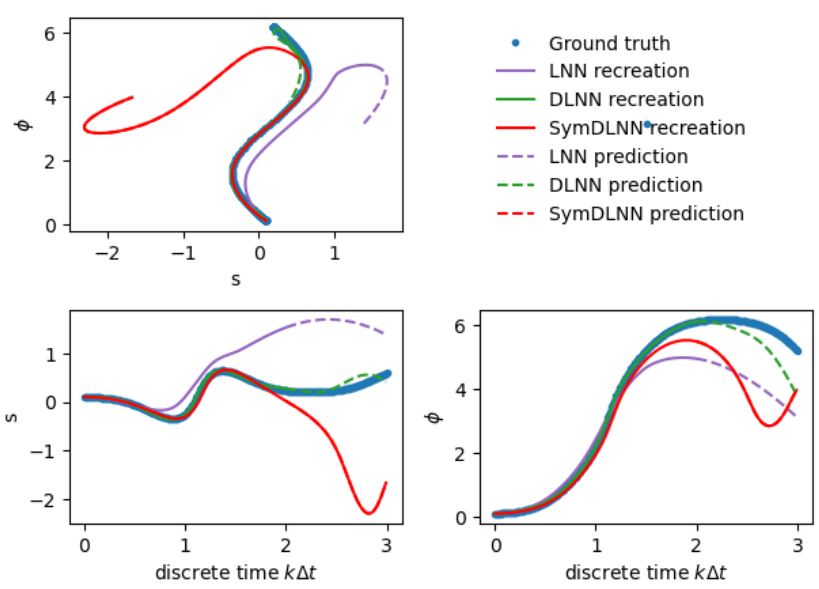
$$w = [0,0],\ M = \left[0, \sqrt{2}/2; -\sqrt{2}/2, 0\right]$$

Reminder : $I(q, \dot{q}) = (M\,q + w)^T\,\nabla_{\dot{q}}\,\mathcal{L}(q, \dot{q})$

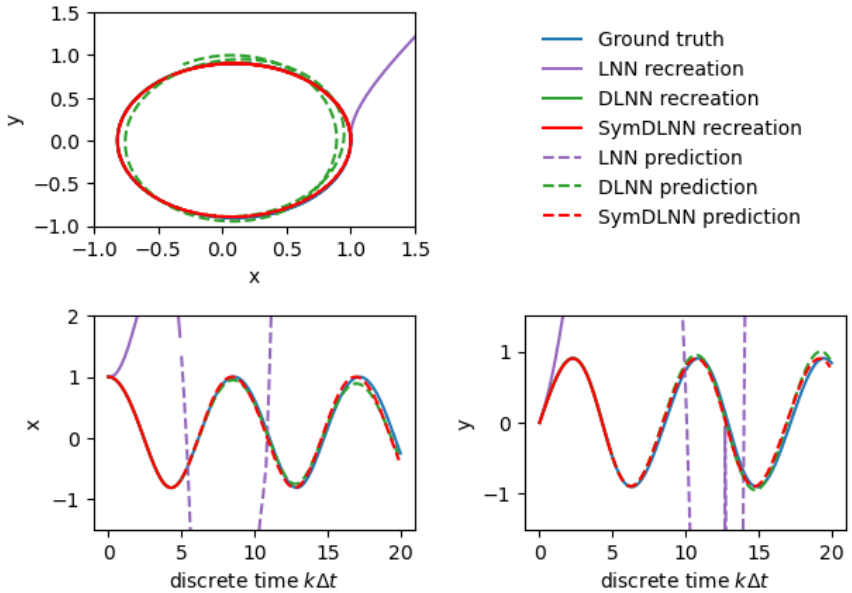
Results

Trajectory recreation and simulation

Pendulum on a cart



Kepler problem



Results

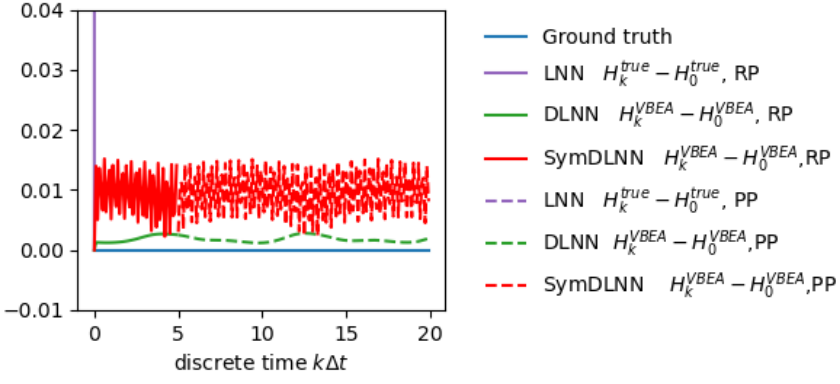
Symmetry and energy recreation and preservation

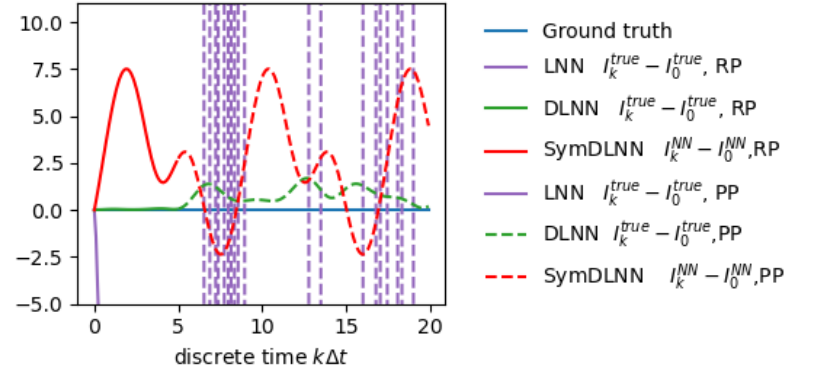
Pendulum on a cart

Original: $\mathrm{M} = [0{,}0; 0{,}0],\ \mathrm{w} = [1{,}0]$

SymDLNN: $\mathrm{w} = [0.997{,}0.082]$,

$$\mathrm{M} = \left[0.00151, -0.00126; -7.83 \times 10^{-6}, -3.62 \times 10^{-5}\right]$$

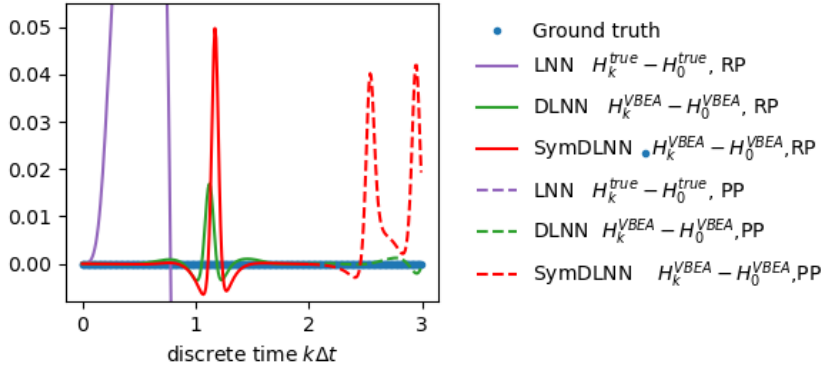


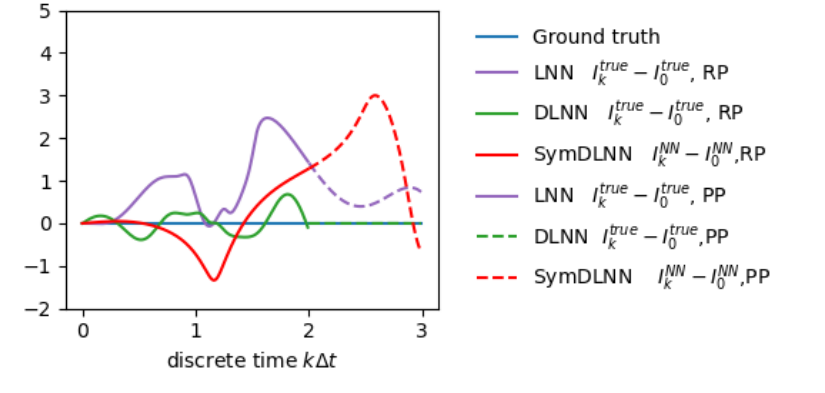


Kepler problem

Original: $\mathrm{M} = [0{,}0.7071; -0.7071{,}0],\ \mathrm{w} = [0{,}0]$

SymDLNN: $\mathrm{M} = [0.074{,}0.813; -0.568, -0.018],\ \mathrm{w} = [0.057{,}0.07]$

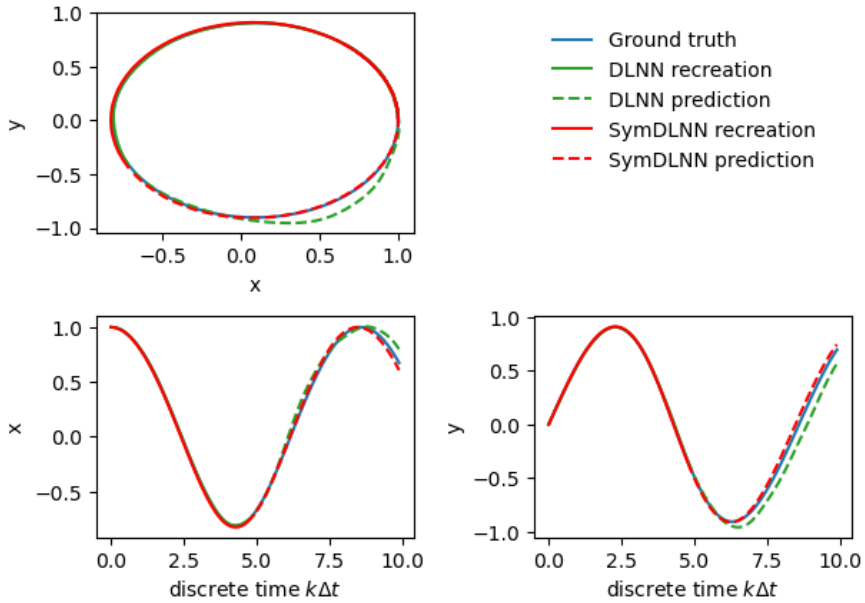




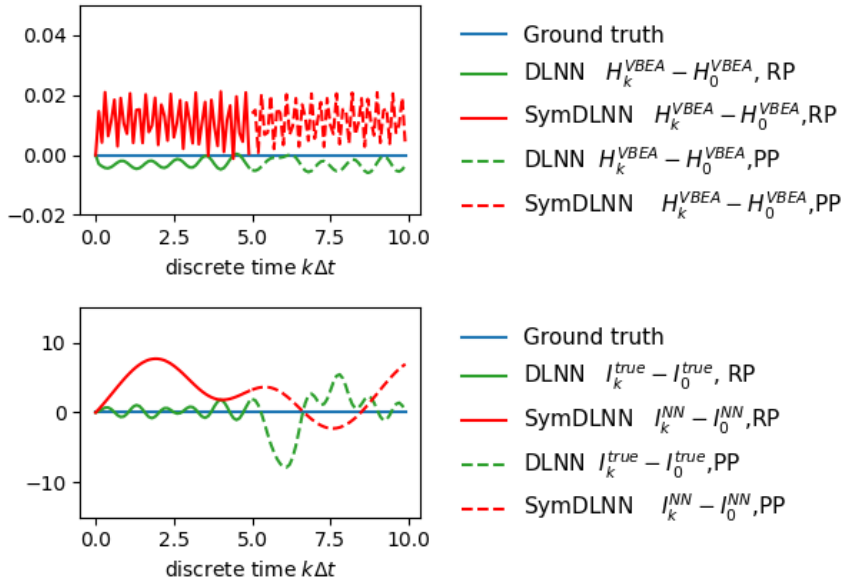
Results

Learning from noisy data- the Kepler problem

Trajectory recreation and prediction



Symmetry and energy preservation



Original: $\mathrm{M} = [0,0.7071; -0.7071,0],\ \mathrm{w} = [0,0]$

SymDLNN: $\mathrm{M} = [0.090,0.812; -0.568,-0.016],\ \mathrm{w} = [0.056,0.077]$

System identification

- **Through learning the equations of motion**
- **Through learning the Lagrangian/Hamiltonian**

with our work we have achieved

- ✓ No restrictions on the expression of the Lagrangian being learnt
- ✓ Only configuration data needed
- ✓ Direct discrete Lagrangian identification -> *reduces the discretisation errors preventing the amplification of the modelling error**
- ✓ Symmetry identification -> qualitative better solution and potentially faster learning
- ✓ Symmetry preserving simulation -> *conservation of energy and/or momenta of the system*
- ✓ Incorporates a degeneracy cost term -> *safeguards against unwanted solutions and against potential numerical issues in forward simulations*
- ✓ Capability to manage noisy data

*** Ober-Blöbaum, S. and Offen, C. (2023).** Variational learning of Euler–Lagrange dynamics from data. Journal of Computational and Applied Mathematics, 421, 114780.

THANK YOU FOR YOUR ATTENTION!

Contact details:

Yana Lishkova, yana.lishkova@eng.ox.ac.uk

Address: St. Hugh's College, St Margaret Road, OX2 6LE, United Kingdom

Main paper:

Lishkova, Y., Scherer, P., Ridderbusch, S., Jamnik, M., Liò, P., Ober-Blöbaum, S. and Offen C. (2022) Discrete Lagrangian Neural Networks with Automatic Symmetry Discovery. arXiv preprint :2211.10830.

Main references:

- **Cranmer, M., Greydanus, S., Hoyer, S., Battaglia, P.,Spergel, D., and Ho, S.** (2020). Lagrangian neural networks. arXiv 2003.04630.
- **Greydanus, S., Dzamba, M., and Yosinski, J.** (2019). Hamiltonian neural networks. Advances in neural information processing systems, 32.
- **Dierkes, E., Offen, C., Ober-Blöbaum, S. and Flaßkamp, K.** (2023). Hamiltonian Neural Networks with Automatic Symmetry Detection, preprint available online
- **Ober-Blöbaum, S. and Offen, C.** (2023). Variational learning of Euler–Lagrange dynamics from data. Journal of Computational and Applied Mathematics, 421, 114780.
- **Iserles, A.,** (2008), A First Course in the Numerical Analysis of Differential Equations. Cambridge Texts in Applied Mathematics. Cambridge University Press, 2 edition, 2008.
- **Razandralandy, D., Hamdouni, A., and Chhay, M.,** (2018), A review of some geometric integrators. Advanced Modeling and Simulation in Engineering Sciences, 5(1):1-67
- **JMarsden, J.E., and West,M.,** "Discrete mechanics and variational integrators," *Acta Numerica*, vol. 10, no. 1, pp. 357–514, 2001